

IEEE WIE ILS 2026

ROUND 2 · PPT & PROTOTYPE VIDEO

National Hackathon

IDEATE. INNOVATE. IMPACT.

POWERING SOLUTIONS. EMPOWERING WOMEN. SHAPING TOMORROW.



SheShield

It acts when she cannot.

A jewellery-style wearable that detects distress automatically, and still lets her cancel a false alarm.



Team Hack Monster'z: we built every layer



Harsh Parmar

Team Lead • Hardware & Firmware

Marwadi University, Rajkot

- Sensor integration: MAX30102 heart-rate + MPU6050 motion
- Confidence-score logic on the ESP32
- Team coordination and problem research



Mirali Sankaliya

SOS Integration & Documentation

Atmiya University, Rajkot

- Cancel-window and SOS messaging logic
- Testing and feedback collection
- Documentation, diagrams and presentation

HOW THE WORK CONNECTS



Marwadi University & Atmiya University, Rajkot, Gujarat • IEEE WIE ILS 2026 National Hackathon • Track 3 – SheLeads
Team Lead: hp259369@gmail.com • 8401357854

Problem: safety devices assume she can act

THE FAILURE POINT

Almost every safety solution needs the user to act: press a button, unlock a phone. In a real assault her hands may be restrained, or she may be in shock.

4,45,256

crimes against women
registered in India in 2022
(about 51 FIRs every hour)

NCRB, Crime in India 2022

32%

of ever-married women (18–49)
reported spousal physical, sexual
or emotional violence

NFHS-5, 2019–21

9 of 12

campus women not confident
they could press a panic button
if restrained or in shock

Our informal conversations, n = 12

7 of 12

had turned off a safety app
because of false alerts

Our informal conversations, n = 12

TODAY'S ASSUMPTION

She is free to act



Presses / unlocks



Alert sent

SHESHIELD'S DIFFERENCE

Distress detected



Confidence score



Alert (can cancel)



Last-mile risk Bus stop to destination: fewer people around, slowest help.



Inclusion Safety is a precondition for women's education, work and public life.

Solution: SheShield detects distress, she stays in control



WHAT IT IS

A jewellery-style wearable (bracelet / pendant concept) that detects distress automatically.
No button press to trigger. She stays in control: press to cancel, long-press for instant SOS.



01 DETECT

Heart-rate + motion, always on

MAX30102 + MPU6050. No button press needed.

Fixes: she cannot press the button



02 DECIDE

Explainable 0-100 confidence score

Heart-rate AND motion must agree. Trigger at 70.

Fixes: false alarms and alert fatigue



03 PROTECT

8-second cancel window, then SOS

Press to cancel. Long-press = instant SOS.

Phone-free in the final product (GSM)

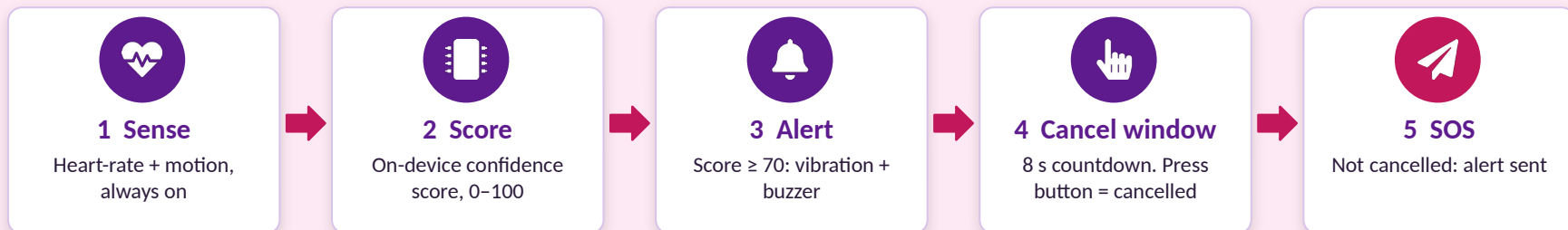
BUILT • PROTOTYPE, shown in our video

- Score, cancel window and long-press SOS working
- SOS via Telegram over hotspot, with live GPS location
- Breadboard build in a project box
- Heart-rate read from the fingertip

PLANNED • FINAL PRODUCT, phone-free

- SOS by GSM SMS: no phone or internet
- Jewellery-style wearable enclosure
- Wrist heart-rate, tuned in a pilot
- Final product est. cost ≈ ₹2,000

How it works: sense, score, cancel, SOS



Signal	Condition	Points
Heart-rate	Rise above her own baseline (10 s calibration): $> +15$ / $> +25$ BPM	+20 / +40
Motion	Struggle-like jerk: medium / high	+20 / +40
Rotation	Violent gyro rotation	+20
Trigger	Score ≥ 70 starts the 8-second countdown	≥ 70

SCORE METER (0-100)



Motion alone: 60 max $\rightarrow$ NO ALERT
Heart-rate and motion must agree.

10 s baseline calibration

70 trigger threshold

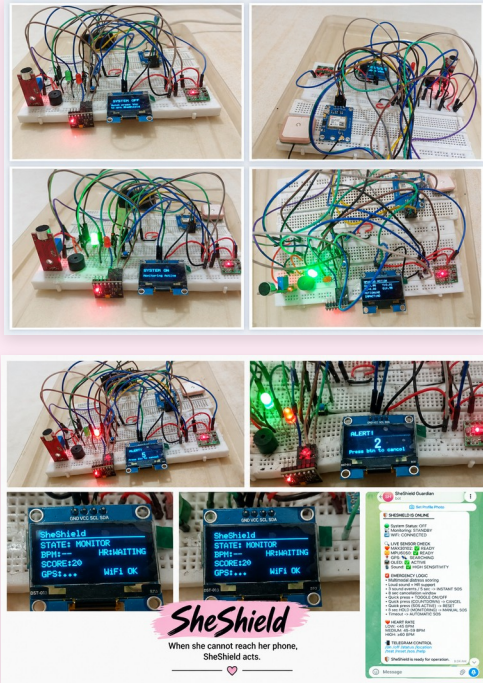
8 s cancel window

2 s long-press SOS

Thresholds are tunable constants set by our own bench tests. No black-box AI. Planned optional 3rd signal: loud sound (MAX9814). SOS today: Telegram via hotspot + live GPS; final: GSM.

Working prototype: full loop on real hardware

✓ WORKING PROTOTYPE



Prototype: OLED states, countdown, Telegram bot

FUNCTIONAL VERIFICATION: DETECTION TO ALERT

✓ Heart-rate sensing

MAX30102

✓ Motion + rotation

MPU6050

✓ Confidence score

0-100, on the ESP32

✓ Automatic trigger

score ≥ 70

✓ 8-second cancel window

button press

✓ Telegram SOS alert

hotspot · live GPS location

✓ 2-second long-press SOS

manual, instant

✓ OLED, buzzer, vibration

live feedback

✓ False-alarm control

motion alone: no alert

WHAT THE PROTOTYPE PROVES

Sense → Score → Trigger → Cancel / Confirm → SOS runs end to end on hardware.

Honest scope Breadboard demo by our team, not a clinical or field study. Alert carries live GPS location (NEO-6M); next: GSM, wrist tuning. Every step is in our video.

Innovation: two-signal score + human cancel window

Typical products	Panic-button pendants	Phone SOS apps	Shake / always-on alerts	SheShield
Needs a conscious press	Yes	Yes (unlock + tap)	Sometimes	No (manual option kept)
Needs her phone	Often	Yes	Often	Final: no. Prototype: hotspot
False-alarm control	None	Low	Low	Score + 8 s cancel window
Explainable logic	Not applicable	Not applicable	Rarely	Yes, rule-based

Columns describe typical product categories, not specific brands.

SINGLE MOST NOVEL ELEMENT

Confidence-scored multi-signal trigger

+

8-second human cancel window

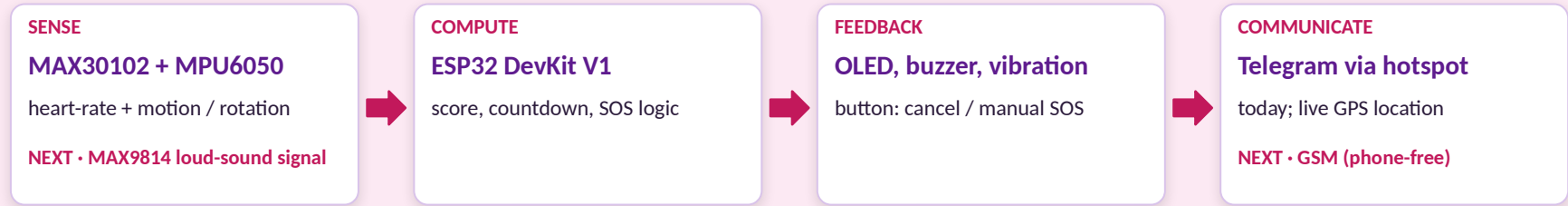
+

Phone-free GSM SOS (final product)

Together they act when she cannot press a button, and keep false alarms low so contacts keep responding.

We do not claim to have invented automatic detection. What is new: passive detection combined with a user-controlled cancel window, in an explainable score.

Technology stack: working core + extended layer



Working prototype part	Role	Approx. cost
ESP32 DevKit V1 (30-pin)	Sensors, score, countdown, Telegram SOS	₹300–450
MAX30102	Heart-rate spike sensing	₹120–200
MPU6050 (GY-521)	Struggle / jerk / rotation	₹80–130
0.96" OLED SSD1306	Live BPM, score, countdown	₹150–200
Buzzer + vibration motor	Alert; 2N2222 / BC547 + 1N4007 driver	₹55
Button, LED, resistors	Cancel / manual SOS, status light	₹25
Breadboard, jumpers, case	Wiring + box (USB power bank: own)	₹150–350

Prices are approximate and vary by shop.

PROTOTYPE COST

≈ ₹900–1,400

SOFTWARE

- Arduino framework (C/C++) + open-source libraries
- Telegram Bot API over HTTPS
- Rule-based score, no cloud AI

EXTENDED LAYER: GPS DONE, NEXT ADD-ONS

- NEO-6M GPS: live location in SOS alert (done)
- MAX9814 / KY-038 (in hand): optional loud-sound signal
- GSM module (planned): phone-free SOS

Feasibility: working prototype today, pilot next



DONE — WORKING PROTOTYPE

- Component selection and breadboard prototype
- Heart-rate and motion pipelines
- Confidence score with baseline calibration
- Cancel window and manual long-press SOS
- OLED display and Telegram SOS
- Live GPS location (NEO-6M) in the SOS alert



NEXT — PLANNED

- GSM module: phone-free SOS
- Compact PCB + wearable enclosure
- Minimal setup app (contacts, sensitivity)
- Closed pilot, 10–15 volunteers + per-user tuning
- Optional loud-sound signal (MAX9814)



TOP TECHNICAL RISK

Wrist heart-rate noise

Mitigation Prototype reads the fingertip. In the pilot: per-user calibration + threshold tuning on the wrist.

FALSE ALARMS

Heart-rate and motion must agree, 8 s cancel, adjustable sensitivity

WEAK INDOOR SIGNAL

Final product: store-and-forward plus manual SOS

PRIVACY

Processing stays on the device until an alert is confirmed

Expected impact: pilot targets, not results

YEAR-1 TARGET — NOT YET ACHIEVED



500–2,000 devices

through college partnerships and IEEE WIE chapters

PILOT KPI TARGETS — GOALS FOR THE PLANNED PILOT, NOT RESULTS

TARGET

$\geq 95\%$

Alert delivery success

TARGET

$\leq 5\%$

False-alarm rate after tuning

TARGET

$< 15\text{ s}$

Trigger to SOS received (GSM version)

TARGET

$\geq 70\%$

Daily wear compliance



INCLUSION

Removes a real barrier to women's education, work and public life.



SUSTAINABILITY

Low-power, repairable, modular hardware with no cloud dependency.

SheShield acts when she cannot.

DETECT · DECIDE · PROTECT



DETECT

Heart-rate + motion, no button press needed



DECIDE

Explainable 0–100 confidence score, trigger at 70



PROTECT

8-second cancel window, then SOS.
Long-press = instant SOS

WORKING TODAY

Prototype ≈ ₹900–1,400, bench-tested, Telegram SOS + live GPS

NEXT

GSM · wearable PCB · pilot, 10–15 volunteers

Our ask Pilot partners (IEEE WIE chapters, colleges) and mentorship to harden the GSM version.

GitHub github.com/HarshParmar029/SheShield-Wearable

Demo video <https://drive.google.com/file/d/1BlSPJShyzV3eIlLE0zheHAcHvunwJRiK/view?usp=sharing>

Thank you · Team Hack Monster'z · Marwadi University & Atmiya University, Rajkot